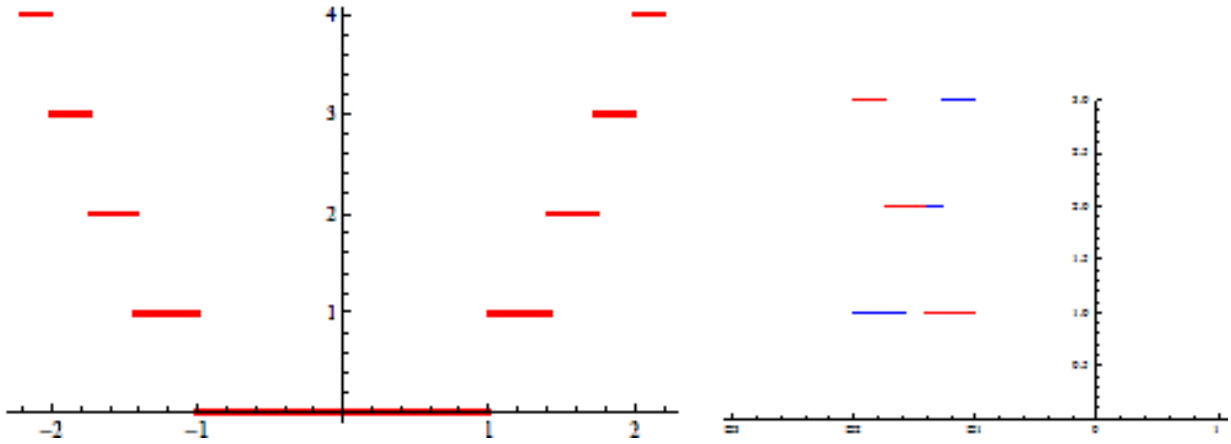
Team Solutions

1. $2^3 3^4 4^5 5^6 = 10^6 \times 2^7 \times 3^4 = 10^6 \times 128 \times 81 \approx 10^6 \times 128 \times 80 = 10^7 \times 1024 = 1.024 \times 10^{10}$ which has $\boxed{11}$ digits.

2. Look at the graph of $y = \lfloor x^2 \rfloor$ and find plateaus of the graphs that are separated by a distance of 3, as $y = \left\lfloor (x+3)^2 \right\rfloor$ is simply the same graph left-shifted by three units. The largest distance between two points on the level $y = 0$ is 2, on the level $y = 1$ is $2\sqrt{2}$, and on the level $y = 2$ is $2\sqrt{3}$. Points in different plateaus on higher levels are separated by a distance greater than three, so all solutions occur on the level $y = 2$. The left interval on the level $y = 2$ of the graph of $y = \lfloor x^2 \rfloor$ is $(-\sqrt{3}, -\sqrt{2}]$, and the right interval on the level $y = 2$ on the graph of $y = \left\lfloor (x+3)^2 \right\rfloor$ is $[\sqrt{2}-3, \sqrt{3}-3)$. The intersection between the two intervals is $[\sqrt{2}-3, -\sqrt{2}]$, so the length of the interval is $\boxed{3 - 2\sqrt{2}}$.



3. For the product to be positive, there must be two or four negative terms in the sequence. Having only two negative terms implies that -3 and -27 are the only terms in the sequence not equal to one, and thus the product is less than 1000. Therefore, the last four terms must all be not equal to one. In that case, the product is at least $-3 \times -27 \times -27 \times -27 > 1000$, so it suffices for all of the terms to be negative. This occurs with probability $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \boxed{\frac{1}{4}}$.

4. $4^4 - 2^2 = 2^8 - 2^2 = 2^2(2^6 - 1)$ and

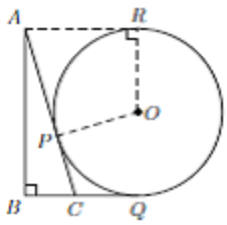
$$\begin{aligned}
&8^8 - 4^4 = 2^{24} - 2^8 = 2^2(2^{22} - 2^6) \\
&= 2^2\left(2^{22} - 2^{16} + 2^{16} - 2^{10} + 2^{10} - 2^4\right) + 2^2\left(2^4 - 2^6\right) \\
&= 2^2\left(2^6 - 1\right)\left(2^{16} + 2^{10} + 2^4\right) + \left(2^6 - 2^8\right) \\
&= 2^2\left(2^6 - 1\right)\left(2^{16} + 2^{10} + 2^4\right) + \left(-2^8 + 2^6 - 2^2 + 2^2\right) \\
&= 2^2\left(2^6 - 1\right)\left(2^{16} + 2^{10} + 2^4\right) - \left(2^8 - 2^2\right) + \left(2^6 - 2^2\right) \\
&= 2^2\left(2^6 - 1\right)\left(2^{16} + 2^{10} + 2^4\right) - 2^2\left(2^6 - 1\right) + \left(2^6 - 2^2\right).
\end{aligned}$$

The first two terms are multiples of $4^4 - 2^2$, so we are only interested in the last term, which is $2^6 - 2^2 = 64 - 4 = \boxed{60}$, which is less than $4^4 - 2^2$.

5. We know that A must be even and non-zero. If A is 6, then $B^C > 0$ so there are $9 \times 10 = 90$ choices for B and C. If A is 4, then B^C must be even and greater than zero, so there are $4 \times 9 = 36$ choices for B and C (as B must be non-zero and even and C must be non-zero). If A is 2 or 8, then B^C must be a multiple of four and greater than zero, so there are $2 \times 9 + 2 \times 8 = 34$ choices for B and C (conditioning on whether B is 0 or 2 mod 4: if B is a multiple of 4 then C must be non-zero, if B is even but not a multiple of four, then C cannot be 0 or 1). The total is 194, so the probability (as a fraction) is $\frac{194}{1000} = \boxed{\frac{97}{500} = 0.194}$.

6. If $b = p \times q$, with p and q primes and $p < q$, then $b!$ is divisible by q^p (as there are exactly p multiples of q between 1 and pq) but not divisible by q^{p+1}, so $(b!)_b$ ends in p zeroes, which won't work, since 10 is not prime. Therefore, $b = p^3$ for a prime p. Then $b!$ is divisible by p^{p^2+p+1} (as there are p^2 multiples of p between 1 and p^3, p multiples of p^2, and 1 multiple of p^3) but not divisible by p^{p^2+p+2}, so $(b!)_b$ ends in $\left\lfloor \frac{p^2+p+1}{3} \right\rfloor$ zeroes. Testing values of p quickly gets $p = 5$, so $b = 5^3 = \boxed{125}$.

7. Let the apex of the cone be A, the center of its base B, and let P be a point of tangency between the side of the cone and one of the spheres. In the vertical cross-section below we label O, Q, and R as the center, bottom, and top of that sphere, and C the point on the base of the cone such that P is on $\overline{AC}$.



If we projected the centers of all three spheres onto the table's surface, B would be the center of an equilateral triangle of side length 2. So $BQ = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$. Note that $AP = AR = BQ$ and $BC + CP = BQ$, so we know that if the radius of the base of the cone is r, then $AP = AR = BC + CQ = r + PC = r + AC - AP \to AC = 2 \times AP - r$:

$$2^2 + r^2 = AC^2 = (2 \times AP - r)^2 = \left(\frac{4\sqrt{3}}{3} - r\right)^2 = r^2 - \frac{8\sqrt{3}}{3}r + \frac{16}{3}$$

$$\frac{8\sqrt{3}}{3}r = \frac{4}{3} \to r = \boxed{\frac{\sqrt{3}}{6}}.$$

8. Invert and subtract 1 from both sides, then take the square root to get

$$|\tan x| = \sqrt{\frac{2\cos^2 22^\circ}{1 - \sin 46^\circ}} = \sqrt{\frac{1 + (2\cos^2 22^\circ - 1)}{1 - \cos 44^\circ}} = \sqrt{\frac{1 + \cos 44^\circ}{1 - \cos 44^\circ}} = \cot 22^\circ.$$

And the smallest positive angle x in degrees that satisfies this is $x = \boxed{68^\circ}$.

9. Using the property $\frac{u}{v}+\frac{x}{y}=\frac{u+x}{v+y}$ if $\frac{u}{v}=\frac{x}{y}$ (unless v, y, or $v+y=0$):

$$\frac{a}{b+1}=3\left(\frac{b-7a}{3b+a}\right)+22\left(\frac{a}{b+1}\right)-\frac{3b+a}{4b+5}=\frac{0}{32+3a+18b}$$

unless $32+3a+18b=0$. If $32+3a+18b\neq 0$, then $a=0$, a contradiction to the conditions in the problem. Therefore $32+3a+18b=0$, so $a+6b=\boxed{-\frac{32}{3}}$.

10. Consider the orbit of 1 under repeated applications of f.

If $f(1)=1$, then the orbit is $1\to 1\to 1\to 1\to 1$, but $1+1+1+1<13$.

If $f(1)=2$, then the next three values must sum to 11, so must be $1\to 2\to 3\to 4\to 4$ or $1\to 2\to 4\to 3\to 4$ ($1\to 2\to 4\to 4\to 3$ is impossible, since 4 maps to two different numbers).

If $f(1)=3$, then the next three values must sum to 10, so must be some arrangement of $4,4,2$ or $4,3,3$, leading to two solutions $1\to 3\to 2\to 4\to 4$ and $1\to 3\to 4\to 2\to 4$. The other possible arrangements lead to contradictions.

If $f(1)=4$, then the next three values must sum to 9, so must be some arrangement of $4,4,1$ or $4,3,2$ or $3,3,3$, leading to three solutions: $1\to 4\to 3\to 3\to 3$, $1\to 4\to 2\to 3\to 4$, and $1\to 4\to 3\to 2\to 4$. The other possible arrangements lead to contradictions. Note that for the first orbit, there are four possible functions with this orbit, since $f(2)$ can be anything in S. The answer is $\boxed{10}$.